ROUND 4 – INTERMEDIATE+ (C++)

1. Star Pattern – Right Angle Triangle

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int rows = 5;
    for(int i=1;i<=rows;i++){
        for(int j=1;j<=rows;j++){
            cout<<"*";
        }
        cout<<endl;
    }
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int rows = 5;
    for(int i=1;i<=rows;i++){
        for(int j=1;j<=i;j++){
            cout<<"*";
        }
        cout<<endl;
    }
    return 0;
}
```

2. Star Pyramid Pattern

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int rows = 5;
    for(int i=1;i<=rows;i++){
        for(int j=1;j<=i;j++){
            cout<<"* ";
        }
        cout<<endl;
    }
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int rows = 5;
    for(int i=1;i<=rows;i++){
        for(int s=1;s<=rows-i;s++){
            cout<<" ";
        }
        for(int j=1;j<=2*i-1;j++){
```

```
            cout<<"*";
        }
        cout<<endl;
    }
    return 0;
}
```

3. Find Missing Number in Array

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,4,5};
    int sum=0;
    for(int i=0;i<4;i++){
        sum+=arr[i];
    }
    cout<<sum;
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,4,5};
    int n=5;
    int expected=n*(n+1)/2;
    int actual=0;
    for(int i=0;i<4;i++){
        actual+=arr[i];
    }
    cout<<expected-actual;
    return 0;
}
```

4. Find Duplicate Elements

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,2,3};
    for(int i=0;i<4;i++){
        cout<<arr[i]<<" ";
    }
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,2,3};
    int n=4;
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
```

```
                cout<<arr[i]<<" ";
                break;
            }
        }
    }
    return 0;
}
```

5. Check Anagram Strings

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    string s1="listen";
    string s2="silent";
    if(s1.length()==s2.length())
        cout<<"Anagram";
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    string s1="listen";
    string s2="silent";
    int count[256]={0};

    if(s1.length()!=s2.length()){
        cout<<"Not Anagram";
        return 0;
    }

    for(int i=0;i<s1.length();i++){
        count[s1[i]]++;
        count[s2[i]]--;
    }

    for(int i=0;i<256;i++){
        if(count[i]!=0){
            cout<<"Not Anagram";
            return 0;
        }
    }

    cout<<"Anagram";
    return 0;
}
```

6. First Non-Repeating Character

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    string s="aabbc";
    cout<<s[0];
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    string s="aabbc";
    int freq[256]={0};

    for(int i=0;i<s.length();i++){
        freq[s[i]]++;
    }

    for(int i=0;i<s.length();i++){
        if(freq[s[i]]==1){
            cout<<s[i];
            break;
        }
    }
    return 0;
}
```

7. Rotate Array Left by K

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,3,4,5};
    int k=2;
    for(int i=k;i<5;i++){
        cout<<arr[i]<<" ";
    }
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[]={1,2,3,4,5};
    int n=5,k=2,temp[5];

    for(int i=0;i<n;i++){
        temp[i]=arr[(i+k)%n];
    }

    for(int i=0;i<n;i++){
        cout<<temp[i]<<" ";
    }
    return 0;
}
```

8. Intersection of Two Arrays

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int a[]={1,2,3};
    for(int i=0;i<3;i++){
```

```
        cout<<a[i]<<" ";
    }
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int a[]={1,2,3};
    int b[]={2,3,4};
    int n=3,m=3;

    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            if(a[i]==b[j]){
                cout<<a[i]<<" ";
                break;
            }
        }
    }
    return 0;
}
```

9. Check Armstrong Number

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
    int n=153,sum=0;
    if(sum==n)
        cout<<"Armstrong";
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    int n=153,temp=n,sum=0;
    while(temp>0){
        int d=temp%10;
        sum+=d*d*d;
        temp/=10;
    }
    if(sum==n)
        cout<<"Armstrong";
    else
        cout<<"Not Armstrong";
    return 0;
}
```

10. Generate All Substrings

Buggy Code:

```
#include <iostream>
using namespace std;

int main() {
```

```
    string s="abc";
    cout<<s;
    return 0;
}
```

Correct Code:

```
#include <iostream>
using namespace std;

int main() {
    string s="abc";
    for(int i=0;i<s.length();i++){
        for(int len=1;len<=s.length()-i;len++){
            cout<<s.substr(i,len)<<endl;
        }
    }
    return 0;
}
```